
Session 1 (2h) — MCU 구조 & 아두이노 회로 구성 기초

Covers: MCU 구조, start of 아두이노 회로 구성 (on-site task)

1. MCU 구조 (MCU structure) — ~40 min

An Arduino Uno is a small computer built around the **ATmega328P microcontroller (MCU)**. Understand these blocks — this is the “MCU 구조” content the matching subject covers:

- **CPU core** — executes your compiled sketch instruction by instruction
- **Flash memory (32KB)** — stores your uploaded program (persists when powered off)
- **SRAM (2KB)** — stores variables while running (lost on power-off/reset)
- **EEPROM (1KB)** — small persistent storage you can read/write from code (not covered in depth this session, but good to know it exists)
- **Clock (16MHz crystal)** — the “heartbeat” that paces every instruction
- **I/O ports** — the physical pins (digital 0–13, analog A0–A5) that connect the MCU to the outside world — this is what Sessions 2–3 dig into

Arduino Uno board layout to know by sight: - USB port (power + programming), barrel jack (external power) - Reset button - Digital pins 0–13 (pins marked ~ support PWM — Session 3) - Analog input pins A0–A5 - Power pins: 5V, 3.3V, GND (multiple), Vin

2. 아두이노 회로 구성 기초 (Circuit configuration basics) — ~50 min

- **Breadboard layout:** outer rails (+/–) run the full length of the board for shared power/ground; middle rows are grouped in 5-hole vertical strips (same strip = same node).
- **Ohm’s Law basics for wiring:** $V = I \times R$ — why an LED always needs a series resistor (typically 220Ω) so it doesn’t draw too much current from a GPIO pin.
- **Golden wiring rules:**
 1. Always connect grounds first when adding a new part to a live layout.
 2. Never connect a GPIO pin directly to 5V/GND without a component (LED, resistor, switch, etc.) in between — that’s a short circuit.
 3. Double-check polarity on LEDs (long leg = anode = toward the pin/resistor).

Practice: [practice/basic_io_test.ino](#) — ~30 min

Wire one LED (D8 → 220Ω resistor → LED → GND) and run the test sketch, which blinks it and prints a startup message over Serial — your first look at both a GPIO output and Serial communication (previewed here, covered properly in Session 5).

Wiring for this session

Component	Pin	Other end
LED (+)	D8	220Ω resistor → anode
LED (–)	GND	cathode

Checkpoint before Session 2

- ☐ Can name the 4 main MCU blocks (CPU, Flash, SRAM, I/O) without looking
- ☐ LED blinks correctly, no short circuits

☐ Can explain why the resistor is required